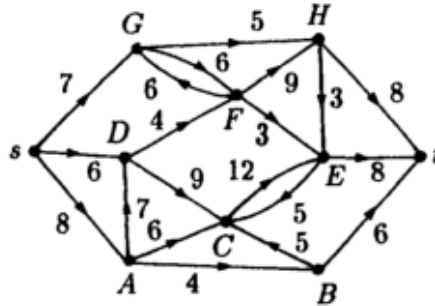


Week 09: Network Flow

1 Exercises

Exercise 1. Construct a maximum flow for the following network (s, A, B, D, E, G, H, t) :



Exercise 2. Given a network $G = (V, E)$ with source s and sink t . The capacities of the edges are given as follows:

- $c(s, A) = 10, c(s, B) = 5$
- $c(A, B) = 2, c(A, C) = 8$
- $c(B, C) = 3, c(B, t) = 4$
- $c(C, t) = 10$

Requirements: Draw the flow network graph for the above parameters. Find the Maximum Flow from s to t . Identify a corresponding Minimum Cut (Min-Cut).

Exercise 3. Given a flow network $G = (V, E)$ with vertex set $V = \{s, v_1, v_2, v_3, v_4, v_5, v_6, t\}$. Edge capacities $c(u, v)$ are defined as follows:

$$\left\{ \begin{array}{l} c(s, v_1) = 14, \quad c(s, v_2) = 11, \\ c(v_1, v_3) = 10, \quad c(v_1, v_4) = 4, \\ c(v_2, v_3) = 0, \quad c(v_2, v_5) = 11, \\ c(v_3, v_6) = 10, \quad c(v_3, v_5) = 0, \\ c(v_4, v_6) = 7, \\ c(v_5, v_4) = 3, \quad c(v_5, t) = 8, \\ c(v_6, t) = 17 \end{array} \right.$$

Requirements: Draw the flow network graph for the above parameters. Find the Maximum Flow from s to t . Identify **all** possible Minimum Cuts (Min-Cuts).

Exercise 4. Suppose that graph G in Question 2 no longer has capacities. Design new capacity values for the edges such that they satisfy all the following requirements:

- There is only one unique minimum (S, T) - cut where $S = \{s, v_1, v_2\}$ and $T = \{v_3, v_4, v_5, v_6, t\}$.
- The maximum flow value is $F_{max} = 15$.

Exercise 5. Given a flow network $G_f = (V, E)$ with vertex set $V = \{s, A, B, t\}$. The current flow state $f(u, v)$ and capacity $c(u, v)$ of the edges are given by the following system:

$$\begin{cases} (s, A) : f = 7, & c = 10 \\ (s, B) : f = 5, & c = 5 \\ (A, B) : f = 0, & c = 2 \\ (A, t) : f = 7, & c = 8 \\ (B, t) : f = 5, & c = 10 \end{cases}$$

Requirements:

- Calculate the residual capacity $c_f(u, v)$ for all pairs of vertices.
- Determine the set of edges E_f of the residual graph G_f or draw the residual graph.
- Find an augmenting path p from s to t on G_f and determine $c_f(p) = \min\{c_f(u, v) : (u, v) \in p\}$.
- Based on the chosen augmenting path, update the graph after performing the flow augmentation.

Exercise 6. Let G be a network containing a pair of opposite edges $u \rightarrow v$ and $v \rightarrow u$ where both have positive capacities. Let G' be a network created from G by reducing the capacities of these edges by

$$\min\{c(u \rightarrow v), c(v \rightarrow u)\}$$

In other words:

- If $c(u \rightarrow v) > c(v \rightarrow u)$, change the capacity of $u \rightarrow v$ to $c(u \rightarrow v) - c(v \rightarrow u)$ and remove edge $v \rightarrow u$.
- If $c(v \rightarrow u) > c(u \rightarrow v)$, change the capacity of $v \rightarrow u$ to $c(v \rightarrow u) - c(u \rightarrow v)$ and remove edge $u \rightarrow v$.
- If $c(u \rightarrow v) = c(v \rightarrow u)$, remove both edges $u \rightarrow v$ and $v \rightarrow u$.

Prove that:

- Every (s, t) maximum flow in G' is also a maximum flow in G . Thus, by simplifying every pair of opposite edges in G , we obtain a more compact network with the same maximum flow value.
- Every minimum cut in G is also a minimum cut in G' and vice versa.
- There is at least one maximum flow in G that is not a maximum flow in G' . Provide an example.

2 Programming Exercises

- Exercise 1.** Write a function that takes an arbitrary weighted graph as input and prints all possible cuts.
- Exercise 2.** Write a function that takes an arbitrary weighted graph, calculates the residual values, and prints the edges of the residual graph with their weights (including flow values and residual capacities).
- Exercise 3.** Implement the Ford-Fulkerson algorithm using DFS to find augmenting paths.
- Exercise 4.** Implement the Edmonds-Karp algorithm to find the shortest augmenting paths.
- Exercise 5.** Given N employees and M jobs. Each employee can only do specific jobs based on their skills. Each employee can do at most x jobs, and each job requires y people. Write a function to assign tasks such that the number of completed jobs is maximized.
- Exercise 6. (k-connected)** For an undirected graph, determine if two vertices s and t are k -connected (there are k edge-disjoint paths between them).
- Exercise 7. (Simple path containing three vertices)** Given an undirected graph and three vertices u, v, w , find a simple path containing all three (or report if none exists).
- Exercise 8. (Unique min-cut)** For an $s - t$ flow network, determine if the minimum cut is unique. *Hint: Solve the min-cut Exercise in G and in G^R .*